
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is foundational to the safe and reliable deployment of machine learning systems in open-world environments, where models are
2 inevitably confronted with inputs drawn from classes wholly absent during training.
3 A promising line of work strengthens OOD detection by leveraging unlabeled
4 in-the-wild data, yet such data is intrinsically difficult to exploit: it arrives as an
5 entangled, unlabeled mixture of in-distribution (InD) and OOD samples, and the
6 absence of any cleanly delineated set of OOD examples obstructs the training of an
7 effective detector. We introduce Medix, a framework that dissolves this difficulty
8 by identifying candidate outliers directly from the unlabeled mixture through the
9 median operation. This reliance on the median is deliberate and principled: as a
10 robust estimator of central tendency, the median is provably insensitive to the noise
11 and OOD contamination that pervade wild data. The outliers it isolates, together
12 with the labeled InD data, then furnish the supervisory signal for training a robust
13 OOD classifier. Theoretically, we derive two-sided, non-asymptotic error bounds
14 proving that Medix attains a low misclassification rate for both inliers and outliers
15 under mild and empirically verifiable assumptions; empirically, our experiments
16 decisively corroborate these guarantees, with Medix consistently and substantially
17 outperforming existing methods across a broad range of open-world settings and
18 thereby confirming our theoretical analysis.
19

20 1 Introduction

21 The deployment of machine learning models in real-world applications inevitably exposes them
22 to out-of-distribution (OOD) inputs, which pose acute challenges for safety and reliability. Such
23 inputs, originating from categories the model was never trained to recognize, must not be assigned
24 confident predictions; yet neural networks are inherently ill-equipped for this task, typically lacking
25 any mechanism by which to detect OOD inputs and appropriately abstain at deployment time (Nguyen
26 et al., 2015). Reliable OOD detection is therefore a prerequisite, rather than an optional refinement,
27 for trustworthy open-world deployment.

28 Identifying OOD samples at inference is at once critical and intrinsically difficult: because models are
29 never exposed to unknown distributions during training, they cannot, unaided, reliably separate OOD
30 from in-distribution (InD) data. To confront this challenge, a recent line of work (Katz-Samuels et al.,
31 2022a; Du et al., 2024a) has turned to auxiliary “in-the-wild” data. In particular, Katz-Samuels et al.
32 (2022a) regularize model training with unlabeled wild data while retaining the objective of classifying
33 labeled InD data. The appeal of wild data is twofold. First, it is readily available—collected essentially
34 for free once a model is operating in its target environment. Second, it lets the model absorb the very
35 distribution of OOD inputs it will encounter at test time, yielding substantially more robust detection.
36 Together these properties make wild data an unusually attractive, low-cost source of supervision for
37 open-world OOD detection.

Exploiting unlabeled wild data, however, is far from straightforward, precisely because such data arrives as an entangled mixture of InD and OOD samples. The absence of any clean, delineated set of OOD examples fundamentally complicates the design of robust detectors: the detector observes only samples drawn from this mixed distribution and is never told whether a given sample is InD or OOD. This regime remains substantially underexplored, leaving considerable room for methodological advancement. Compounding the gap, few studies furnish a formal theoretical foundation; to the best of our knowledge, Du et al. (2024a) stands as the sole prior work to provide such a foundation for the “in-the-wild” setting, underscoring the pressing need for principled, guarantee-bearing treatments of this problem.

In parallel, median-based estimation has recently proven highly effective in data pruning (Acharya et al., 2024), owing to its intrinsic robustness against corrupted or atypical samples. These two threads—the open problem of exploiting unlabeled wild data and the demonstrated robustness of median-based estimation—converge naturally, and together motivate the central question this paper sets out to answer:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question decisively in the affirmative by introducing a median-centric perspective on OOD detection. Concretely, we formulate a median-based optimization framework and develop an accompanying algorithm, Medix, that identifies OOD samples within wild data at a demonstrably low error rate. We establish a theoretical foundation that guarantees this low error and validate it empirically, establishing both the robustness and the efficiency of the algorithm. In doing so, we contribute one of the few rigorous theoretical foundations for the unlabeled “in-the-wild” setting. Specifically, we prove that median-based filtering is provably robust for outlier detection in unlabeled mixtures by upper-bounding the fraction of InD samples erroneously flagged as outliers. This fraction is governed by two complementary and interpretable effects: a *contamination effect*, which quantifies the influence of OOD points and provably remains controlled whenever the OOD proportion stays below 50%; and a *concentration effect*, which exploits the sub-Gaussian behavior of InD gradients to bound the deviation of individual InD points from their mean gradient. Acting jointly, these two effects guarantee that, with high probability, the number of misclassified InD points remains small—certifying the reliability of the method even under substantial, near-worst-case contamination.

We benchmark Medix against two families of methods: (1) those trained solely on InD data, and (2) those trained on both InD data and an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves markedly over the strong baseline KNN+ (Sun et al., 2022), lowering average FPR95 by 40.98%. Crucially, whereas approaches such as Outlier Exposure (Hendrycks et al., 2019) depend on a clean auxiliary dataset—imposing the stringent distributional assumption that the auxiliary data be fully separable from the InD data—Medix attains superior results while dispensing with any such assumption, and is therefore applicable across a substantially broader range of realistic deployment settings. Relative to the state-of-the-art wild-data method WOODS (Katz-Samuels et al., 2022a), Medix further reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10, establishing a consistent advantage over the strongest competing approach in both regimes.

We summarize our key contributions as follows:

- C1) A median-centric filtering framework.** We propose Medix, a median-centric greedy method that first filters outliers from unlabeled wild data and then trains an OOD detector on the extracted outliers together with the labeled InD samples. The gradient-based median filtering stage is our central methodological contribution: it is precisely what enables robust OOD detection without recourse to a clean auxiliary set, and it thereby lifts the restrictive separability assumption on which prior regularization-based detectors depend.
- C2) Two-sided theoretical guarantees.** We establish provable, non-asymptotic two-sided guarantees for the robustness of median-based filtering in jointly identifying inliers and outliers within unlabeled mixtures. Concretely, we prove that the misclassification rates for both InD samples flagged as outliers and OOD samples retained as inliers are tightly controlled by two interpretable effects: a contamination effect, which remains provably bounded whenever the OOD proportion is below 50%, and a concentration effect, which follows from the stable, sub-Gaussian gradient behavior of InD samples. To our knowledge,

91 this constitutes one of the few rigorous foundations available for the unlabeled in-the-wild
 92 setting.

93 **C3) Comprehensive empirical validation.** We conduct an extensive evaluation of Medix across
 94 eleven InD–OOD pairs and against 20 competitive baselines, and Medix outperforms every
 95 one of them, delivering uniformly superior detection across the board. We further show that
 96 the Medix outlier-extraction stage attains a strikingly low error rate (e.g., only 12.5%; see
 97 Figure 2), directly and quantitatively corroborating the guarantees of C2.

98 2 Preliminaries and Problem Setup

99 In this section we formalize the OOD detection problem and then specify precisely the data setup,
 100 model architecture, loss functions, and learning objective that underpin the remainder of the paper.

101 **Labeled In-Distribution Data.** Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{1, \dots, K\}$ the label
 102 space; jointly, these two spaces characterize the structure of the InD data. A labeled dataset $\mathcal{S}_{\text{in}} =$
 103 $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ comprises n pairs drawn independently and identically distributed (i.i.d.)
 104 from $\mathbb{P}_{XY}$, an unknown joint distribution over $\mathcal{X} \times \mathcal{Y}$. We write $\mathbb{P}_{\text{in}}$ for the marginal of $\mathbb{P}_{XY}$ on
 105 $\mathcal{X}$ —that is, the underlying distribution of InD inputs—and train the InD model on $\mathcal{S}_{\text{in}}$.

106 **Out-of-distribution detection.** We consider the practical scenario in which a model trained on
 107 labeled InD data is subsequently deployed in environments that may present OOD inputs from classes
 108 unseen during training, i.e., inputs carrying some label $y \notin \mathcal{Y}$. On such inputs the model must
 109 abstain rather than commit to a class prediction. The central objective at inference time is therefore
 110 to decide whether a given input is drawn from the InD distribution or instead originates from an
 111 OOD source—a binary decision that must remain reliable across the diverse, unanticipated inputs
 112 characteristic of open-world deployment.

113 **Unlabeled wild data.** A principal obstacle in OOD detection is the scarcity of labeled OOD samples:
 114 the OOD sample space is potentially unbounded, rendering the collection of labeled examples both
 115 costly and impractical. To circumvent this obstacle, and following Katz-Samuels et al. (2022a),
 116 we incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into our learning framework, thereby
 117 faithfully reflecting the conditions of real-world deployment. Wild data constitutes a blend of InD
 118 and OOD samples and is readily gathered, at negligible cost, once a model pre-trained on $\mathcal{S}_{\text{in}}$ is put
 119 into operation. Consistent with Du et al. (2024a); Katz-Samuels et al. (2022a), we adopt the Huber
 120 contamination model (Huber, 1964) to characterize the marginal distribution of the wild data,

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

121 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. The boundary
 122 case $\pi = 0$ corresponds to the complete absence of OOD samples and renders the problem trivial; we
 123 therefore confine attention to the informative regime $\pi \in (0, 1]$.

124 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier parameterized by $\phi \in \Phi$,
 125 with Φ its parameter space; the output of f_ϕ is a soft probability distribution over the $K = |\mathcal{Y}|$ InD
 126 classes, and its supervised loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection we introduce a separate
 127 classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$, parameterized by $\theta \in \Theta$ over parameter space Θ , with associated binary loss
 128 $\ell_b(g_\theta(x), y_b)$ and $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. Here the positive value $y_+ > 0$ encodes the InD class and
 129 the negative value $y_- < 0$ the OOD class, so that the sign of g_θ directly yields the detection decision
 130 at inference.

131 **Learning objective.** Our framework jointly trains the OOD detector g_θ and the multi-class classifier
 132 f_ϕ , drawing simultaneously on the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. At test time we assess
 133 performance with the following complementary metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

134 where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ is the OOD test distribution. FPR and TPR jointly
 135 characterize the quality of OOD detection, while $\text{Acc}(f_\phi)$ certifies that InD classification accuracy is
 136 preserved rather than traded away for detection performance—a dual criterion under which a method
 137 must succeed on both axes simultaneously.

3 Method: Median-Centric Framework for OOD Detection

We now present **Medix**, a learning paradigm that performs OOD detection by systematically harnessing unlabeled wild data. By construction, Medix overcomes the limitations of conventional approaches confined to InD data, and it is especially well-suited to open-world deployment, where models are routinely confronted with previously unseen inputs. The framework decomposes into two tightly coupled stages: **1) Outlier Extraction**, a filtering procedure that isolates candidate OOD samples from the unlabeled wild data (Section 3.1); and **2) Detector Training**, in which a binary OOD detector is trained on the InD data together with the outlier candidates identified in stage 1 (Section 3.2), following the protocol of Du et al. (2024a). As the following sections establish, this two-stage design not only extracts OOD data from the unlabeled wild set effectively, but also furnishes a robust, theoretically grounded foundation for deploying machine learning models in dynamic, open-world environments—the outlier-extraction stage being the locus of both our methodological novelty and our theoretical guarantees.

3.1 Extracting Candidate Outliers from the Wild Data

To isolate candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix adopts an optimization-based formulation grounded in the parameter gradients of a classification model f_ϕ trained exclusively on the InD dataset $\mathcal{S}_{\text{in}}$. The underlying rationale is precise: the gradient of the InD loss at a given sample, taken with respect to the parameters of an InD-trained model, measures how far that sample departs from InD behavior, and therefore constitutes a discriminative signal for filtering. We formalize this procedure below.

Reference gradient estimation from InD data. The first step establishes a reference gradient from the InD dataset $\mathcal{S}_{\text{in}}$. We train the classifier f_ϕ on $\mathcal{S}_{\text{in}}$ via empirical risk minimization (ERM),

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. With the classifier trained, we define the reference gradient $\bar{\nabla}_{\text{in}}$ as the empirical mean, over the InD data, of the per-sample loss gradients with respect to the model parameters:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

The reference gradient $\bar{\nabla}_{\text{in}}$ thus serves as a stable anchor in gradient space against which the deviation of any other data point can be quantified.

Motivation. The design of Medix rests on a precise and directly testable hypothesis: as the proportion of OOD samples in the wild dataset $\mathcal{S}_{\text{wild}}$ increases, the aggregate gradient statistic of $\mathcal{S}_{\text{wild}}$ should deviate progressively from the reference InD gradient $\bar{\nabla}_{\text{in}}$. To test this hypothesis directly, we conduct a controlled experiment with CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. We initialize $\mathcal{S}_{\text{wild}}$ with 10,000 CIFAR-10 samples held strictly disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ (and hence to compute $\bar{\nabla}_{\text{in}}$), and then incrementally inject SVHN OOD samples while tracking the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild-set gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

As Figure 1 shows, the L_2 -norm deviation increases clearly and monotonically with the OOD count, unambiguously confirming our hypothesis. This monotonic relationship is far more than a diagnostic observation: it directly motivates the algorithm developed in the next section and, in particular, its stopping criterion, under which the procedure terminates once the inter-iteration change in L_2 -norm deviation falls below a threshold ϵ —a mechanism we detail below.

Filtering potential outliers from unlabeled wild data. Building on the monotonic behavior established in Figure 1, we cast outlier identification as the following optimization problem, whose solution yields the outlier subset $\mathcal{S}_{\text{out}}^*$ within $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function and $\hat{y}_{\tilde{x}_i}$ the predicted label of a wild sample $\tilde{x}_i$; for notational brevity we write $G_S = \{\nabla \ell(\tilde{x}_i)\}_{i \in S}$. Intuitively, (4) seeks the subset $\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}$ whose gradient EWM lies closest to the reference gradient $\bar{\nabla}_{\text{in}}$. Given the monotonic relationship established in Figure 1, the minimizer $\mathcal{S}_{\text{in}}^*$ recovers precisely the InD portion of $\mathcal{S}_{\text{wild}}$, so that its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ isolates the OOD portion.

Because exhaustively solving (4) is combinatorial, and hence computationally prohibitive as $|\mathcal{S}_{\text{wild}}|$ grows, we introduce a tractable greedy approximation based on a leave-one-out criterion, formalized in Algorithm 1. The algorithm performs iterative outlier detection by quantifying, at each step, how the removal of an individual wild sample reshapes the gradient EWM relative to the InD reference.

At each iteration the algorithm computes the EWM of the current wild-set gradients and compares it against the reference gradient $\bar{\nabla}_{\text{in}}$. Let $d_t = \|\text{EWM}(G_S) - \bar{\nabla}_{\text{in}}\|$ denote the L_2 distance between $\bar{\nabla}_{\text{in}}$ and the EWM of the gradients remaining in the wild set $\mathcal{S}$. For each candidate $i \in \mathcal{S}$, the algorithm evaluates the leave-one-out EWM $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ together with the resulting distance $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$, thereby obtaining the induced reduction $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$; the k samples that induce the largest reductions—and hence contribute most to the deviation from the InD reference—are flagged as OOD and removed. This process iterates until no remaining sample yields a significant reduction δ_i , or until a maximum iteration count is reached. Formally, the convergence criterion requires the inter-iteration change in L_2 distance to fall below a predefined threshold ϵ . This criterion follows directly from the monotonic trend observed in Figure 1: once a substantial fraction of OOD samples has been removed, the L_2 distance settles to a small, stable value, precisely marking the point at which the algorithm should halt—thereby avoiding the erroneous excision of genuine InD samples that would degrade downstream performance.

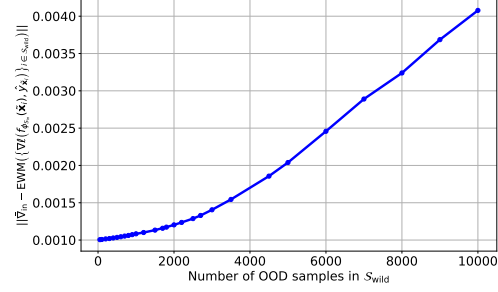


Figure 1: The L_2 -norm deviation between the reference InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM of the wild-set gradients increases monotonically as OOD samples are progressively added to $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD, Wide ResNet). This clean, monotonic trend is precisely the empirical basis for both the filtering objective in (4) and the convergence criterion of Algorithm 1: it certifies that the EWM tracks the OOD proportion and that halting on a stabilized deviation isolates the InD subset.

Algorithm 1 Iterative Outlier Detection via Medix

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{x}_i), \hat{y}_{\tilde{x}_i}})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{x}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_S) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

3.2 Training the OOD Detector with Candidate Outliers

Having extracted the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we train the OOD detector g_θ to maximally separate InD data from the extracted candidate outliers, following the protocol of Du et al. (2024a). The objective enforces separability at the decision boundary (thresholded at 0), driving g_θ toward positive outputs on labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs on candidate outliers

219 $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. Concretely, the loss decomposes into a positive InD term and a negative outlier term,

$$\begin{aligned}\mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}.\end{aligned}\tag{5}$$

220 Because the 0/1 loss is non-differentiable, we replace it with a differentiable sigmoid-based binary
221 surrogate, which preserves alignment with the original objective while rendering the optimization
222 tractable under gradient-based methods. The framework simultaneously retains the InD risk of
223 Equation 2, thereby safeguarding predictive accuracy on InD samples even as detection is optimized.
224 This unified objective substantially strengthens the generalization of g_θ and enables robust detection
225 of OOD samples drawn from $\mathbb{P}_{\text{out}}$, as our experiments in Section 5 confirm.

226 4 Theoretical Analysis

227 We now formalize the theoretical guarantees underlying the filtering stage of Medix. The two
228 theorems below furnish provable, non-asymptotic upper bounds on the misclassification rates for InD
229 and OOD points, respectively, and, taken together, they establish a complete two-sided robustness
230 guarantee for EWM filtering: neither are too many inliers discarded nor too many outliers retained.
231 Complete proofs are deferred to Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

232 Theorem 4.1 establishes that the inlier misclassification rate is governed by exactly two effects: a
233 *concentration term*, which captures statistical fluctuations in the InD gradients and vanishes as the
234 sample size grows, and a *contamination term*, which quantifies the influence of OOD points on the
235 EWM. Decisively, the bound remains controlled for every contamination ratio $\pi < 0.5$ —a property
236 that directly certifies the robustness of median-based filtering even under substantial OOD presence.
237 This naturally raises the complementary question: *how many true outliers evade the filter and are*
238 *mistakenly retained as inliers?* The following theorem settles this question by bounding the OOD
239 misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

240 Theorem 4.2 complements the preceding result by bounding the fraction of OOD points mistakenly
241 retained as InD. Together, Theorems 4.1 and 4.2 constitute a complete two-sided robustness guarantee
242 for Medix filtering, under which the inlier and outlier misclassification rates are jointly governed by
243 the balance among three interpretable effects:

244 **Contamination Effect.** In Theorem 4.1, the median-based criterion remains provably stable even
245 when the wild dataset contains a large fraction π of OOD points, provided only that $\pi < 0.5$. This

behavior is encoded by the term $\pi/[2(1 - \pi)]$, which grows with π and precisely captures the risk that OOD gradients skew the EWM. Symmetrically, Theorem 4.2 carries the mirror-image penalty $\frac{1-\pi}{2\pi}$, which quantifies the reciprocal difficulty of isolating OOD points when they constitute the underrepresented population.

Concentration Effect. Both bounds exploit the sub-Gaussian structure of the gradient coordinates. For InD data, this structure guarantees that the bulk of gradients concentrate tightly about $\bar{\nabla}_{\text{in}}$, keeping the median stable even at finite sample sizes; for OOD data, concentration about μ_{out} lets us quantify the risk that a misaligned OOD sample slips past the filter. Both concentration terms decay at the parametric rate $1/\sqrt{m}$, with m the number of samples from the corresponding class, and therefore vanish in the large-sample limit.

Separation Effect. Unique to Theorem 4.2, the separation effect specifies how far the OOD mean gradient must lie from the InD mean gradient for OOD samples to be reliably rejected. The exponential factor $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ encodes this trade-off exactly: the greater the separation between the two gradient distributions, the exponentially smaller the probability that an OOD gradient deceives the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering achieves robust separation of InD and OOD samples in unlabeled mixtures. The fraction of InD samples misclassified as outliers is bounded by the contamination and concentration effects, while the fraction of OOD points incorrectly retained is bounded by an exponential separation term, a concentration term, and a reverse contamination effect. Together, these results furnish rigorous theoretical assurance that Medix simultaneously controls both error types under mild, empirically verifiable assumptions—a two-sided guarantee that, to our knowledge, has no counterpart among existing wild-data detectors.

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The coordinate-wise sub-Gaussianity of InD gradients is the linchpin of our concentration bounds and, more broadly, a standard cornerstone of robust statistical estimation. This assumption is strongly supported by the empirical evidence. Figure 4a shows that the histogram of InD gradient values is bell-shaped and tightly concentrated about the mean, exhibiting precisely the light, exponentially decaying tails that characterize sub-Gaussian random variables and that render extreme gradient values rare. Figure 4b corroborates this conclusion through a Q-Q plot (Wilk and Gnanadesikan, 1968) that compares the empirical gradient quantiles against the theoretical Gaussian quantiles: the near-exact alignment with the 45-degree reference line indicates that the empirical distribution closely tracks a Gaussian, and since Gaussians are a subclass of sub-Gaussian variables, the assumption is directly and empirically justified. *Moreover, we establish a more general variant of these bounds that dispenses with the sub-Gaussian assumption entirely, requiring only bounded second and fourth moments; this appears as Theorem C.3 (Appendix C.3). Although the rates degrade gracefully in this weaker regime, the core robustness guarantee of Medix persists intact, demonstrating that our conclusions extend well beyond the sub-Gaussian setting.*

5 Experiments

This section establishes the efficacy of Medix across a broad range of InD–OOD dataset pairs, benchmarking it comprehensively against 20 widely used baselines. All experiments are conducted on NVIDIA A100-SXM4-80GB GPUs, and the code required to reproduce our results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. To ensure a rigorous and fair comparison with prior work, we adopt the exact experimental protocol of Katz-Samuels et al. (2022a), the study that introduced the problem of learning OOD detectors from wild data. We use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$), and for OOD testing we employ a diverse suite of natural-image datasets: PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) ($\mathbb{P}_{\text{out}}$). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) at a default contamination level $\pi = 0.5$ —the most demanding balanced-mixture regime admitted by our theory,

and therefore a deliberately stringent test of robustness. Concretely, when PLACES365 serves as the OOD test set, we form the wild mixture by combining CIFAR with PLACES365 and evaluate on PLACES365; this procedure is repeated identically for every OOD dataset and every baseline, ensuring strict comparability. Each InD CIFAR set is partitioned into two disjoint halves: the first 25,000 images train $\phi_{S_{in}}$, while the remainder form the wild mixture S_{wild} . Gradients are computed on the penultimate-layer weights, which prior work has established to be particularly informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard, complementary metrics: (1) the False Positive Rate (FPR95 $\downarrow$) of OOD samples at a 95% True Positive Rate on InD samples, (2) the Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy (InD Acc $\uparrow$). The first two quantify the quality of OOD detection, while the third verifies that InD classification performance is preserved rather than traded away—so that improvements in detection are not obtained at the expense of the underlying classifier.

Baselines. Our comparison spans a broad and competitive set of baselines, systematically organized by whether they are trained on InD data alone or on both InD and wild data. Among methods trained exclusively on InD data ($\mathbb{P}_{in}$), we compare Medix against a comprehensive range of established detectors: Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all built on a softmax cross-entropy-trained model. For completeness we further include the contrastive-loss methods CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Among methods that leverage both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training by promoting lower confidence or higher energy on outlier data; against WOODS (Katz-Samuels et al., 2022a), which introduced the notion of wild unlabeled data and exploits it through constrained optimization; and against the more recent CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), included to ensure a thorough and up-to-date evaluation against the current state of the art.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), we instantiate the InD classifier $\phi_{S_{in}}$ as a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2, trained by stochastic gradient descent with momentum 0.9, weight decay 0.0005, and initial learning rate 0.1 for 100 epochs under cosine learning-rate decay, with batch size 128 and dropout rate 0.3. The Medix hyperparameters ϵ and k are selected from $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$ respectively, scaled to dataset size so as to maximize OOD detection performance. The OOD classifier g_θ is initialized from the pre-trained InD classifier $\phi_{S_{in}}$ and augmented with a linear layer that performs binary classification on the penultimate-layer features. It is then fine-tuned for 100 epochs at learning rate 0.001 according to Equation (5), which combines the binary classification loss with the InD classification loss at a relative weight of 10 on the binary term; all remaining training parameters are held identical to those used for $\phi_{S_{in}}$.

5.3 Results

Our principal results on CIFAR-100 appear in Table 2, where Medix outperforms every OOD detection baseline by a decisive margin. Two patterns stand out. First, methods that exploit both InD and wild data substantially surpass those confined to InD data: relative to the strong InD-only baseline KNN+, Medix lowers FPR95 by 52.31% on PLACES365 and by 38.24% on TEXTURES, precisely quantifying the substantial value of in-the-wild data as a regularizing signal. Second, and more tellingly, Medix also dominates the strongest wild-data competitors: it attains an average FPR95 of 5.42% on CIFAR-100—a 1.32% absolute improvement over the state-of-the-art WOODS—while sustaining a competitive InD accuracy of 73.33%. The modest accuracy gap is fully and transparently accounted for by the experimental protocol: Medix is trained on 25,000 labeled InD samples, whereas the InD-only baselines use the full 50,000-sample CIFAR-100 training set. It is therefore a direct consequence of the wild-data setting rather than any limitation intrinsic to the method, and it is more than offset by the pronounced gains in detection. The CIFAR-10 results in Table 1 tell an entirely consistent story: Medix surpasses all baselines, improving FPR95 over WOODS by 7.21%

Table 1: OOD detection performance of Medix versus all baselines with CIFAR-10 as InD data, reported per OOD test set (SVHN, PLACES365, LSUN-C, LSUN-Resize, TEXTURES) and as an average, alongside InD accuracy. All numbers are averaged over five runs (Medix reported with standard deviations); the best result in each column is in **bold**, with lower FPR95 and higher AUROC being better. Medix attains the best FPR95 and the best AUROC on *every* OOD test set, surpassing both the InD-only detectors and the strongest wild-data baselines—including WOODS—while retaining competitive InD accuracy; its average FPR95 of 0.80% is more than four times lower than that of the best competing method.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance of Medix versus all baselines with CIFAR-100 as InD data, reported per OOD test set and as an average, alongside InD accuracy. All numbers are averaged over five runs (Medix reported with standard deviations); the best result in each column is in **bold**, with lower FPR95 and higher AUROC being better. On this more challenging 100-class benchmark, Medix again attains the best FPR95 and the best AUROC on every OOD test set, decisively outperforming even the strongest wild-data baselines and lowering the average FPR95 to 5.42% against the 6.74% of the state-of-the-art WOODS.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using $\mathbb{P}_{in}$ only												
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
	Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

on PLACES365 and 5.25% on TEXTURES, thereby confirming that its effectiveness at detecting OOD samples holds uniformly across distinct benchmarks rather than being specific to any single dataset.

A representative visual example of Medix. To render the behavior of Algorithm 1 (Outlier Extraction) directly interpretable, we construct a deliberately simple two-dimensional synthetic experiment. The InD data is drawn from three multivariate Gaussians—one per class—with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$ and shared covariance $0.25 \cdot I$, while the OOD data follows $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 shows, Medix recovers the outliers with near-perfect fidelity: it correctly flags 87.5% of the true OOD samples, and the extracted set incurs only 12.5% InD contamination. This concrete instance directly and vividly substantiates the low extraction error established by our theoretical analysis, translating the two-sided bounds of Section 4 into a visible, quantitative outcome.

Additional studies and insights. Owing to space constraints, we defer a substantial body of further experiments to Appendix A, each of which independently reinforces the conclusions above: (1) hyperparameter-selection ablations and a sensitivity analysis (Appendix A.2) establishing the strong robustness of Medix to its hyperparameters; (2) a comparison of EWM against the geometric median

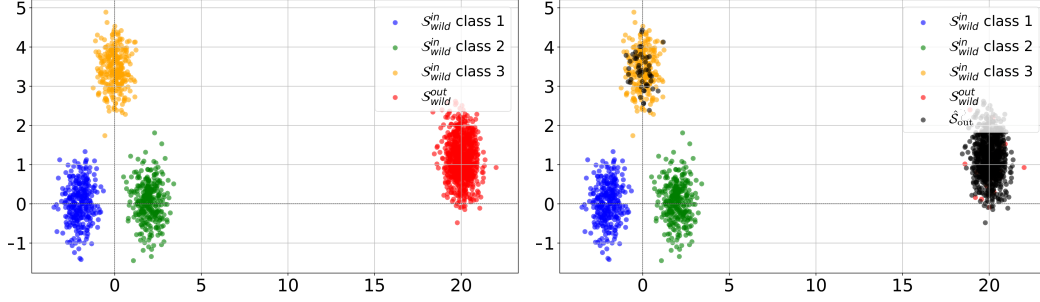


Figure 2: Medix applied to a synthetic 2D wild mixture. (a) The wild set comprises InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$, drawn from three multivariate Gaussians, together with OOD data $\mathcal{S}_{\text{wild}}^{\text{out}}$. (b) The outliers $\hat{\mathcal{S}}_{\text{out}}$ flagged by Medix (black) align almost exactly with the ground-truth OOD points: only 12.5% of $\hat{\mathcal{S}}_{\text{out}}$ are in fact InD points, furnishing direct visual confirmation of the low extraction error predicted by Theorems 4.1 and 4.2.

(Appendix A.1), which demonstrates that EWM is more sensitive to distributional shift and is therefore the more effective and reliable filtering choice; (3) a comprehensive comparison against methods trained with competitive contrastive objectives (Appendix A.3), in which Medix outperforms even these strong baselines by a significant margin; (4) an evaluation on large-scale data under the demanding unseen-OOD setting $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4), where Medix again leads by a wide margin; (5) an assessment of the computation and memory efficiency of Medix (Appendix A.6); (6) an analysis of the impact of pseudo-label quality (Appendix A.5), which demonstrates resilience to noisy or low-confidence labels; and (7) a comparison with semi-supervised open-set recognition methods (Appendix A.7). Collectively, these studies confirm that the advantages of Medix are robust across model architectures, data scales, and evaluation regimes.

6 Related Work

OOD detection has attracted intense and sustained interest in recent years (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022c; Wang et al., 2022c; Galil et al., 2023; Djurisić et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One prominent family derives scoring functions to gauge the distributional origin of an input, spanning distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020). Effective as they are, these post-hoc scores operate on a fixed InD-trained model and, unlike Medix, do not exploit unlabeled wild data during training.

A second family imposes regularization during training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a), for instance by suppressing model confidence (Lee et al., 2017; Hendrycks et al., 2019) or raising its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. Critically, however, most of these methods presuppose access to a clean auxiliary OOD dataset—a restrictive assumption that Medix dispenses with outright.

A related line of work relaxes the labeled-only assumption by incorporating unlabeled wild data (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023; Katz-Samuels et al., 2022a; Geng et al., 2025), yet these studies stop short of furnishing an explicit outlier-detection mechanism. Du et al. (2024a,b) do introduce explicit outlier filtering, but their thresholding technique differs fundamentally from ours: Medix instead detects outliers through a novel median-centric criterion equipped with

provable guarantees. A further, structural distinction is equally consequential. Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, in which each batch contains a fixed ratio of InD and OOD samples; with large outsourced datasets such structure is simply unavailable, since data is mixed randomly across the corpus. Medix removes this requirement entirely, operating directly and without modification under the more realistic regime of dataset-level mixing. Finally, many works adopt positive-unlabeled learning, training classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022); the essential difference is that these methods only separate $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and do not simultaneously train an OOD classifier. In sharp contrast, Medix couples a median-centric outlier-identification procedure—with provably low error rates—directly to the training of a dedicated OOD detector.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection that transforms unlabeled in-the-wild data into a reliable training signal. By exploiting the inherent robustness of the median operation, Medix filters outliers from mixed unlabeled data and thereby enables the training of a dependable OOD detector without any clean auxiliary set. Our theoretical analysis furnished provable, non-asymptotic bounds on the inlier misclassification rate, establishing that Medix remains robust even under substantial OOD contamination—up to 50%—with errors controlled by sub-Gaussian concentration and contamination effects, and complemented these with matching bounds on the OOD misclassification rate that guarantee accurate isolation of outliers under clear separation. Empirically, Medix surpassed all 20 baselines across diverse benchmarks: it reduced average FPR95 by 40.98% relative to strong baselines such as KNN+, decisively outperformed state-of-the-art wild-data methods including WOODS and DRL, and attained an outlier-extraction error rate as low as 12.5%. Taken together, these tightly coupled theoretical and empirical results establish median-centric filtering as a principled, robust, and highly effective foundation for OOD detection from unlabeled data, and open a clear path toward reliable open-world deployment.

References

- Abbas, M., Azmat, M., Horesh, R., and Yurochkin, M. (2025). Out-of-distribution detection using synthetic data generation. In *Second Conference on Language Modeling*.
- Acharya, A., Dhillon, I. S., and Sanghavi, S. (2024). Geometric median (gm) matching for robust data pruning. *arXiv preprint arXiv:2406.17188*.
- Acharya, A., Sanghavi, S., Jing, L., Bhushanam, B., Choudhary, D., Rabbat, M., and Dhillon, I. (2022). Positive unlabeled contrastive learning. *arXiv preprint arXiv:2206.01206*.
- Bai, H., Canal, G., Du, X., Kwon, J., Nowak, R. D., and Li, Y. (2023). Feed two birds with one scone: Exploiting wild data for both out-of-distribution generalization and detection. In *International Conference on Machine Learning*, pages 1454–1471. PMLR.
- Bendale, A. and Boulton, T. E. (2016). Towards open set deep networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1563–1572.
- Chapel, L., Alaya, M. Z., and Gasso, G. (2020). Partial optimal transport with applications on positive-unlabeled learning. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 2903–2913. Curran Associates, Inc.
- Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. (2014). Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3606–3613.
- Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2022). Extremely simple activation shaping for out-of-distribution detection. *arXiv preprint arXiv:2209.09858*.
- Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2023). Extremely simple activation shaping for out-of-distribution detection. In *The Eleventh International Conference on Learning Representations*.
- Du, X., Fang, Z., Diakonikolas, I., and Li, Y. (2024a). How does unlabeled data provably help out-of-distribution detection? In *The Twelfth International Conference on Learning Representations*.

445 Du, X., Ghosh, R., Sim, R., Salem, A., Carvalho, V., Lawton, E., Li, Y., and Stokes, J. W. (2024b). Vlmguard:
446 Defending vlms against malicious prompts via unlabeled data. *arXiv preprint arXiv:2410.00296*.

447 Du, X., Gozum, G., Ming, Y., and Li, Y. (2022a). Siren: Shaping representations for detecting out-of-distribution
448 objects. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in*
449 *Neural Information Processing Systems*, volume 35, pages 20434–20449. Curran Associates, Inc.

450 Du, X., Sun, Y., Zhu, J., and Li, Y. (2023). Dream the impossible: Outlier imagination with diffusion models.
451 In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural*
452 *Information Processing Systems*, volume 36, pages 60878–60901. Curran Associates, Inc.

453 Du, X., Wang, X., Gozum, G., and Li, Y. (2022b). Unknown-aware object detection: Learning what you don’t
454 know from videos in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern*
455 *Recognition*, pages 13678–13688.

456 Du, X., Wang, Z., Cai, M., and Li, S. (2022c). Towards unknown-aware learning with virtual outlier synthesis.
457 In *International Conference on Learning Representations*.

458 Fan, Y., Kukleva, A., Dai, D., and Schiele, B. (2023). Ssb: Simple but strong baseline for boosting performance
459 of open-set semi-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer*
460 *Vision (ICCV)*, pages 16068–16078.

461 Fang, Z., Li, Y., Lu, J., Dong, J., Han, B., and Liu, F. (2022). Is out-of-distribution detection learnable? In
462 Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural*
463 *Information Processing Systems*, volume 35, pages 37199–37213. Curran Associates, Inc.

464 Fort, S., Ren, J., and Lakshminarayanan, B. (2021). Exploring the limits of out-of-distribution detection.
465 *Advances in Neural Information Processing Systems*, 34:7068–7081.

466 Gal, Y. and Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty
467 in deep learning. In Balcan, M. F. and Weinberger, K. Q., editors, *Proceedings of The 33rd International*
468 *Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–
469 1059, New York, New York, USA. PMLR.

470 Galil, I., Dabbah, M., and El-Yaniv, R. (2023). A framework for benchmarking class-out-of-distribution detection
471 and its application to imagenet. In *The Eleventh International Conference on Learning Representations*.

472 Garg, S., Balakrishnan, S., and Lipton, Z. (2022). Domain adaptation under open set label shift. In Koyejo,
473 S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information*
474 *Processing Systems*, volume 35, pages 22531–22546. Curran Associates, Inc.

475 Garg, S., Wu, Y., Smola, A. J., Balakrishnan, S., and Lipton, Z. (2021). Mixture proportion estimation and
476 pu learning: a modern approach. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
477 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8532–8544. Curran
478 Associates, Inc.

479 Geifman, Y. and El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. In
480 Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine*
481 *Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR.

482 Geng, C., Li, Q., Wang, X., Liang, D., Chen, S., and Yuen, P. C. (2025). Lod: Loss-difference ood detection by
483 intentionally label-noisifying unlabeled wild data. *arXiv preprint arXiv:2505.12952*.

484 Ghosal, S. S., Sun, Y., and Li, Y. (2024). How to overcome curse-of-dimensionality for out-of-distribution
485 detection? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19849–19857.

486 Gong, T., Wang, G., Ye, J., Xu, Z., and Lin, M. (2018). Margin based pu learning. In *Proceedings of the AAAI*
487 *Conference on Artificial Intelligence*, volume 32.

488 Hang, J.-Y. and Zhang, M.-L. (2024). Binary decomposition: A problem transformation perspective for open-set
489 semi-supervised learning. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and
490 Berkenkamp, F., editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235
491 of *Proceedings of Machine Learning Research*, pages 17505–17518. PMLR.

492 He, R., Li, R., Han, Z., Yang, X., and Yin, Y. (2023). Topological structure learning for weakly-supervised
493 out-of-distribution detection. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages
494 4858–4866.

495 Hein, M., Andriushchenko, M., and Bitterwolf, J. (2019). Why relu networks yield high-confidence predictions
496 far away from the training data and how to mitigate the problem. In *Proceedings of the IEEE/CVF conference*
497 *on computer vision and pattern recognition*, pages 41–50.

498 Hendrycks, D. and Gimpel, K. (2016). A baseline for detecting misclassified and out-of-distribution examples in
499 neural networks. *arXiv preprint arXiv:1610.02136*.

500 Hendrycks, D. and Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in
501 neural networks. In *International Conference on Learning Representations*.

502 Hendrycks, D., Mazeika, M., and Dietterich, T. (2019). Deep anomaly detection with outlier exposure. In
503 *International Conference on Learning Representations*.

504 Hsieh, C.-J., Natarajan, N., and Dhillon, I. (2015). Pu learning for matrix completion. In Bach, F. and Blei, D.,
505 editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings*
506 *of Machine Learning Research*, pages 2445–2453, Lille, France. PMLR.

507 Huang, R., Geng, A., and Li, Y. (2021a). On the importance of gradients for detecting distributional shifts in the
508 wild. *Advances in Neural Information Processing Systems*, 34:677–689.

509 Huang, R., Geng, A., and Li, Y. (2021b). On the importance of gradients for detecting distributional shifts in the
510 wild. In Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information*
511 *Processing Systems*.

512 Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–
513 101.

514 Jeong, T. and Kim, H. (2020). Ood-maml: Meta-learning for few-shot out-of-distribution detection and
515 classification. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
516 *Neural Information Processing Systems*, volume 33, pages 3907–3916. Curran Associates, Inc.

517 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022a). Training ood detectors in their natural habitats.
518 In *International Conference on Machine Learning*, pages 10848–10865. PMLR.

519 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022b). Training OOD detectors in their natural
520 habitats. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings*
521 *of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning*
522 *Research*, pages 10848–10865. PMLR.

523 Kim, S., Kim, D., Cho, M., and Kwak, S. (2020). Proxy anchor loss for deep metric learning. In *2020 IEEE/CVF*
524 *Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3235–3244.

525 Kristiadi, A., Hein, M., and Hennig, P. (2020). Being bayesian, even just a bit, fixes overconfidence in relu
526 networks. In *International conference on machine learning*, pages 5436–5446. PMLR.

527 Krizhevsky, A., Hinton, G., et al. (2009). Learning multiple layers of features from tiny images.

528 Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation
529 using deep ensembles. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S.,
530 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates,
531 Inc.

532 Lee, K., Lee, H., Lee, K., and Shin, J. (2017). Training confidence-calibrated classifiers for detecting out-of-
533 distribution samples. *arXiv preprint arXiv:1711.09325*.

534 Lee, K., Lee, K., Lee, H., and Shin, J. (2018a). A simple unified framework for detecting out-of-distribution
535 samples and adversarial attacks. *Advances in neural information processing systems*, 31.

536 Lee, K., Lee, K., Lee, H., and Shin, J. (2018b). A simple unified framework for detecting out-of-distribution
537 samples and adversarial attacks. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N.,
538 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates,
539 Inc.

540 Letouzey, F., Denis, F., and Gilleron, R. (2000). Learning from positive and unlabeled examples. In *International*
541 *Conference on Algorithmic Learning Theory*, pages 71–85. Springer.

542 Liang, S., Li, Y., and Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in
543 neural networks. In *International Conference on Learning Representations*.

544 Liu, J., Lin, Z., Padhy, S., Tran, D., Bedrax Weiss, T., and Lakshminarayanan, B. (2020a). Simple and principled
545 uncertainty estimation with deterministic deep learning via distance awareness. In Larochelle, H., Ranzato,
546 M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*,
547 volume 33, pages 7498–7512. Curran Associates, Inc.

548 Liu, W., Wang, X., Owens, J., and Li, Y. (2020b). Energy-based out-of-distribution detection. *Advances in*
549 *neural information processing systems*, 33:21464–21475.

550 Liu, W., Wang, X., Owens, J., and Li, Y. (2020c). Energy-based out-of-distribution detection. In Larochelle,
551 H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing*
552 *Systems*, volume 33, pages 21464–21475. Curran Associates, Inc.

553 Maddox, W. J., Izmailov, P., Garipov, T., Vetrov, D. P., and Wilson, A. G. (2019). A simple baseline for bayesian
554 uncertainty in deep learning. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and
555 Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

556 Malinin, A. and Gales, M. (2018). Predictive uncertainty estimation via prior networks. In Bengio, S., Wallach,
557 H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R., editors, *Advances in Neural Information*
558 *Processing Systems*, volume 31. Curran Associates, Inc.

559 Malinin, A. and Gales, M. (2019). Reverse kl-divergence training of prior networks: Improved uncertainty and
560 adversarial robustness. In Wallach, H., Larochelle, H., Beygelzimer, A., d'Alché-Buc, F., Fox, E., and Garnett,
561 R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

562 Meinke, A. and Hein, M. (2020). Towards neural networks that provably know when they don't know. In
563 *International Conference on Learning Representations*.

564 Ming, Y., Fan, Y., and Li, Y. (2022). POEM: Out-of-distribution detection with posterior sampling. In
565 Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th*
566 *International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*,
567 pages 15650–15665. PMLR.

568 Ming, Y. and Li, Y. (2024). How does fine-tuning impact out-of-distribution detection for vision-language
569 models? *International Journal of Computer Vision*, 132(2):596–609.

570 Ming, Y., Sun, Y., Dia, O., and Li, Y. (2023). How to exploit hyperspherical embeddings for out-of-distribution
571 detection? In *The Eleventh International Conference on Learning Representations*.

572 Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. (2011). Reading digits in natural images
573 with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*,
574 volume 2011, page 4. Granada.

575 Nguyen, A., Yosinski, J., and Clune, J. (2015). Deep neural networks are easily fooled: High confidence
576 predictions for unrecognizable images. In *Proceedings of the IEEE conference on computer vision and pattern*
577 *recognition*, pages 427–436.

578 Niu, G., Du Plessis, M. C., Sakai, T., Ma, Y., and Sugiyama, M. (2016). Theoretical comparisons of positive-
579 unlabeled learning against positive-negative learning. *Advances in neural information processing systems*,
580 29.

581 Peng, B., Luo, Y., Zhang, Y., Li, Y., and Fang, Z. (2024). Conjnorm: Tractable density estimation for out-of-
582 distribution detection. In *The Twelfth International Conference on Learning Representations*.

583 Plessis, M. D., Niu, G., and Sugiyama, M. (2015). Convex formulation for learning from positive and unlabeled
584 data. In Bach, F. and Blei, D., editors, *Proceedings of the 32nd International Conference on Machine Learning*,
585 volume 37 of *Proceedings of Machine Learning Research*, pages 1386–1394, Lille, France. PMLR.

586 Ren, J., Fort, S., Liu, J., Roy, A. G., Padhy, S., and Lakshminarayanan, B. (2021). A simple fix to mahalanobis
587 distance for improving near-ood detection. *arXiv preprint arXiv:2106.09022*.

588 Ren, J., Luo, J., Zhao, Y., Krishna, K., Saleh, M., Lakshminarayanan, B., and Liu, P. J. (2022). Out-of-distribution
589 detection and selective generation for conditional language models. In *The Eleventh International Conference*
590 *on Learning Representations*.

591 Saito, K., Kim, D., and Saenko, K. (2021). Openmatch: Open-set semi-supervised learning with open-set
592 consistency regularization. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W.,
593 editors, *Advances in Neural Information Processing Systems*, volume 34, pages 25956–25967. Curran
594 Associates, Inc.

- 595 Sehswag, V., Chiang, M., and Mittal, P. (2021). {SSD}: A unified framework for self-supervised outlier detection.
596 In *International Conference on Learning Representations*.
- 597 Sun, Y., Guo, C., and Li, Y. (2021). React: Out-of-distribution detection with rectified activations. *Advances in*
598 *Neural Information Processing Systems*, 34:144–157.
- 599 Sun, Y. and Li, Y. (2022). Dice: Leveraging sparsification for out-of-distribution detection. In *European*
600 *Conference on Computer Vision*, pages 691–708. Springer.
- 601 Sun, Y., Ming, Y., Zhu, X., and Li, Y. (2022). Out-of-distribution detection with deep nearest neighbors. In
602 *International Conference on Machine Learning*, pages 20827–20840. PMLR.
- 603 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020a). Csi: Novelty detection via contrastive learning on distributionally
604 shifted instances. *Advances in neural information processing systems*, 33:11839–11852.
- 605 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020b). Csi: Novelty detection via contrastive learning on distributionally
606 shifted instances. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
607 *Neural Information Processing Systems*, volume 33, pages 11839–11852. Curran Associates, Inc.
- 608 Tao, L., Du, X., Zhu, J., and Li, Y. (2023). Non-parametric outlier synthesis. In *The Eleventh International*
609 *Conference on Learning Representations*.
- 610 Uppaal, R., Hu, J., and Li, Y. (2023). Is fine-tuning needed? pre-trained language models are near perfect
611 for out-of-domain detection. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the*
612 *61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages
613 12813–12832, Toronto, Canada. Association for Computational Linguistics.
- 614 Van Amersfoort, J., Smith, L., Teh, Y. W., and Gal, Y. (2020). Uncertainty estimation using a single deep
615 deterministic neural network. In *International conference on machine learning*, pages 9690–9700. PMLR.
- 616 Wang, H., Li, Z., Feng, L., and Zhang, W. (2022a). Vim: Out-of-distribution with virtual-logit matching. In
617 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4921–4930.
- 618 Wang, H., Liu, W., Bocchieri, A., and Li, Y. (2021). Can multi-label classification networks know what they
619 don’t know? *Advances in Neural Information Processing Systems*, 34:29074–29087.
- 620 Wang, Q., Fang, Z., Zhang, Y., Liu, F., Li, Y., and Han, B. (2023a). Learning to augment distributions for
621 out-of-distribution detection. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine,
622 S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 73274–73286. Curran
623 Associates, Inc.
- 624 Wang, Q., Liu, F., Zhang, Y., Zhang, J., Gong, C., Liu, T., and Han, B. (2022b). Watermarking for out-of-
625 distribution detection. *Advances in Neural Information Processing Systems*, 35:15545–15557.
- 626 Wang, Q., Ye, J., Liu, F., Dai, Q., Kalander, M., Liu, T., HAO, J., and Han, B. (2023b). Out-of-distribution
627 detection with implicit outlier transformation. In *The Eleventh International Conference on Learning*
628 *Representations*.
- 629 Wang, Y., Zou, J., Lin, J., Ling, Q., Pan, Y., Yao, T., and Mei, T. (2022c). Out-of-distribution detection via
630 conditional kernel independence model. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K.,
631 and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 36411–36425.
632 Curran Associates, Inc.
- 633 Wang, Z., Xiang, L., Huang, L., Mao, J., Xiao, L., and Yamasaki, T. (2024). Scomatch: Alleviating overtrusting
634 in open-set semi-supervised learning. In *European Conference on Computer Vision*, pages 217–233. Springer.
- 635 Wei, H., Xie, R., Cheng, H., Feng, L., An, B., and Li, Y. (2022). Mitigating neural network overconfidence
636 with logit normalization. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S.,
637 editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings*
638 *of Machine Learning Research*, pages 23631–23644. PMLR.
- 639 Wen, Y., Tran, D., and Ba, J. (2020). Batchensemble: an alternative approach to efficient ensemble and lifelong
640 learning. In *International Conference on Learning Representations*.
- 641 Wilk, M. B. and Gnanadesikan, R. (1968). Probability plotting methods for the analysis of data. *Biometrika*,
642 55(1):1–17.
- 643 Wu, Q., Chen, Y., Yang, C., and Yan, J. (2023). Energy-based out-of-distribution detection for graph neural
644 networks. In *The Eleventh International Conference on Learning Representations*.

645 Xu, D. and Denil, M. (2021). Positive-unlabeled reward learning. In *Conference on Robot Learning*, pages
646 205–219. PMLR.

647 Yang, J., Wang, H., Feng, L., Yan, X., Zheng, H., Zhang, W., and Liu, Z. (2021). Semantically coherent
648 out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*,
649 pages 8301–8309.

650 Yang, J., Wang, P., Zou, D., Zhou, Z., Ding, K., Peng, W., Wang, H., Chen, G., Li, B., Sun, Y., et al.
651 (2022). Openood: Benchmarking generalized out-of-distribution detection. *Advances in Neural Information
652 Processing Systems*, 35:32598–32611.

653 Yang, J., Zhou, K., Li, Y., and Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International
654 Journal of Computer Vision*, 132(12):5635–5662.

655 Yu, F., Seff, A., Zhang, Y., Song, S., Funkhouser, T., and Xiao, J. (2015). Lsun: Construction of a large-scale
656 image dataset using deep learning with humans in the loop. *arXiv preprint arXiv:1506.03365*.

657 Zagoruyko, S. (2016). Wide residual networks. *arXiv preprint arXiv:1605.07146*.

658 Zhang, J., Yang, J., Wang, P., Wang, H., Lin, Y., Zhang, H., Sun, Y., Du, X., Zhou, K., Zhang, W., et al. (2023).
659 Openood v1.5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*.

660 Zhang, Y., Lu, J., Peng, B., Fang, Z., and ming Cheung, Y. (2024). Learning to shape in-distribution feature space
661 for out-of-distribution detection. In *The Thirty-eighth Annual Conference on Neural Information Processing
662 Systems*.

663 Zhao, Y., Xu, Q., Jiang, Y., Wen, P., and Huang, Q. (2022). Dist-pu: Positive-unlabeled learning from a label
664 distribution perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern
665 Recognition*, pages 14461–14470.

666 Zheng, H., Wang, Q., Fang, Z., Xia, X., Liu, F., Liu, T., and Han, B. (2023). Out-of-distribution detection
667 learning with unreliable out-of-distribution sources. *Advances in Neural Information Processing Systems*,
668 36:72110–72123.

669 Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., and Torralba, A. (2017). Places: A 10 million image database for
670 scene recognition. *IEEE transactions on pattern analysis and machine intelligence*, 40(6):1452–1464.

671 Zhou, Z., Guo, L.-Z., Cheng, Z., Li, Y.-F., and Pu, S. (2021). Step: Out-of-distribution detection in the presence
672 of limited in-distribution labeled data. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
673 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29168–29180. Curran
674 Associates, Inc.

675 Zhu, J., Geng, Y., Yao, J., Liu, T., Niu, G., Sugiyama, M., and Han, B. (2023). Diversified outlier exposure
676 for out-of-distribution detection via informative extrapolation. *Advances in Neural Information Processing
677 Systems*, 36:22702–22734.

678 Zhu, Y., Chen, Y., Xie, C., Li, X., Zhang, R., Xue, H., Tian, X., zheng, b., and Chen, Y. (2022). Boosting
679 out-of-distribution detection with typical features. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave,
680 D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages
681 20758–20769. Curran Associates, Inc.

A Additional Studies

A.1 Element-wise Median vs Geometric Median

A natural question is why Medix employs the element-wise median rather than the geometric median (Acharya et al., 2024) for outlier detection. We settle it with a direct, controlled comparison. To this end, we ran a controlled experiment with CIFAR-10 as the InD dataset and SVHN as the OOD dataset. Here $\mathcal{S}_{\text{wild}}$ consists of 5k CIFAR-10 samples held strictly disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ (and hence to compute $\bar{\nabla}_{\text{in}}$); we then incrementally add SVHN OOD samples while jointly tracking the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the wild-set gradients, together with the proportion of OOD samples removed. As Figure 3 demonstrates, at any matched OOD count the element-wise median flags a substantially higher proportion of OOD samples as outliers than the geometric median. This establishes conclusively that the element-wise median is markedly more sensitive to distributional shift, making it the more effective and reliable filtering criterion for Medix.

A.2 Hyperparameter Selection and Sensitivity Analysis

We conducted an ablation to quantify precisely the sensitivity of Medix to the hyperparameters ϵ and k , using CIFAR-100 as the InD dataset and SVHN as the OOD dataset. As Table 3 shows, Medix is strongly robust to variation in both ϵ and k . It attains a remarkably low FPR95 of 0.16 at $\epsilon = 0.005$, $k = 20000$, and—crucially—perturbing these settings degrades performance only gracefully rather than catastrophically: reducing ϵ to 0.0005 or halving k to 10000 yields FPR95 values of 0.20 and 0.68 respectively, each of which still comfortably surpasses every baseline. Thus, while careful tuning yields the strongest results, Medix retains its decisive advantage across a wide range of hyperparameter choices. This insensitivity to exact tuning is a substantial practical asset, rendering Medix a dependable choice for real-world deployments in which exhaustive hyperparameter search is infeasible.

Table 3: Effect of the hyperparameters ϵ and k on Medix OOD detection (CIFAR-100 InD, SVHN OOD, FPR95 $\downarrow$), with representative InD-only and wild-data baselines listed for reference. Across every tested (ϵ, k) configuration, Medix remains far below all baselines, confirming that its advantage does not hinge on precise tuning.

Method	FPR95 $\downarrow$	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.3 Comprehensive comparison with recent competitive methods

We extend our comparison to a suite of competitive baselines built on diverse contrastive objectives—SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al.

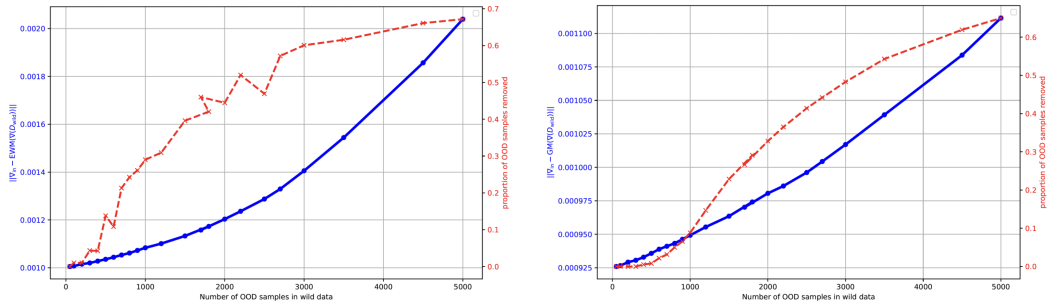


Figure 3: Element-wise median (EWM) versus geometric median (GM) as OOD samples are progressively added to $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD, Wide ResNet): at any matched OOD count, EWM flags a substantially higher proportion of true OOD samples as outliers, directly evidencing its greater sensitivity to distributional shift and thereby motivating its adoption as the filtering criterion of Medix.

Table 4: OOD detection performance of Medix versus recent competitive baselines with CIFAR-100 as InD data (ResNet-34), reported per OOD test set and as an average. All numbers are averaged over five runs (Medix reported with standard deviations); the best result in each column is in **bold**, with lower FPR95 and higher AUROC being better. Medix leads on every single OOD test set, decisively outperforming even the strong methods trained with contrastive objectives and improving the average FPR95 by more than 27 points over the best competing baseline.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

(2023)—which we include for completeness even though Medix does not itself employ contrastive learning. We further add the more recent CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c) to ensure a thorough and up-to-date evaluation. For this study we use CIFAR-100 as the InD dataset and train the ResNet-34 model¹ following Ming et al. (2023); Zhang et al. (2024): stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs, initial learning rate 0.5 under cosine scheduling, and batch size 512, reusing the checkpoints of Ming et al. (2023)². As Table 4 shows, Medix outperforms every one of these competitive baselines by a significant margin—including those trained with contrastive objectives—attaining an average FPR95 of 8.68% and an average AUROC of 97.76%. Beyond this decisive gain in raw performance, Medix uniquely furnishes a provably low error rate, offering rigorous theoretical guarantees that the great majority of these alternatives lack.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We now examine whether Medix scales to large-scale wild OOD data under the demanding unseen-OOD setting. This regime is challenging on two distinct counts: (1) it draws on a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), which sharply increases both data complexity and computational demand; and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately rendered distributionally distinct from the training-time wild OOD data, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, thereby faithfully mirroring realistic deployment in which the OOD encountered at test time need not coincide with what was observed during training. We take CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD data. The substantial distributional gap between the large-scale wild OOD data (300K Random Images) and the OOD test data (SVHN) makes this a particularly stringent test of generalization to unseen outliers. Comparing against baselines that likewise leverage wild data during training, Table 5 shows that Medix is decisively superior, delivering both a markedly lower FPR95 (41.29 ± 1.2)

Table 5: OOD detection performance on large-scale unseen OOD data (CIFAR-100 InD, 300K Random Images as wild OOD, SVHN as test OOD), compared against wild-data baselines. Even under this stringent distributional-gap setting, Medix roughly halves the FPR of the best competing baseline while simultaneously attaining the highest AUROC—evidence that its advantage persists when the test-time OOD distribution is entirely unseen during training.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

¹Since ResNet-34 is the standard architecture across these works, we adopt the same model to ensure consistency and to facilitate a strictly fair comparison.

²<https://github.com>

and a higher AUROC (87.25 ± 0.6)—compelling evidence that it separates InD from OOD data effectively even when the test-time OOD distribution is entirely unseen during training.

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

A further question concerns the role of pseudo-label quality $\hat{y}_{\tilde{x}_i}$: does the softmax confidence of the pseudo-labels materially affect OOD filtering, and could a simple pre-filtering step that discards low-confidence pseudo-labels improve robustness? To settle this question directly, we evaluate the effect of removing low-confidence pseudo-labels, using CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. We compute the softmax probability of each pseudo-labeled sample and discard predictions falling below a confidence threshold of 0.6, which removes 15.98% of the samples from the training set.

Performance is essentially unchanged by this intervention: both variants yield an identical FPR95 of 0.01%, and the filtering step nudges AUROC only marginally, from 99.98% to 99.99%. Removing low-confidence pseudo-labels therefore has no meaningful effect on either robustness or OOD detection, which demonstrates conclusively that Medix is resilient to noisy or low-confidence pseudo-labels and that additional pre-filtering is unlikely to yield appreciable gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next address whether the Medix filtering phase is computationally feasible and memory-efficient at scale. To this end we profiled the procedure on an NVIDIA A100-SXM4-80GB GPU with roughly 15,000 unlabeled samples per InD–OOD pair, using CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As Table 6 reports, GPU memory usage remains modest—below 100 MB throughout—and the filtering phase completes within a tractable wall-clock budget even for thousands of unlabeled samples. Medix filtering is therefore computationally practical and imposes no excessive memory overhead, confirming its suitability for deployment on large datasets.

Table 6: Profiling of the Medix filtering phase on an NVIDIA A100 80GB GPU ($\approx 15,000$ unlabeled samples per pair): wall-clock time together with peak and current GPU memory for two InD–OOD pairs. Peak memory overhead remains under 100 MB for both pairs, confirming the practical scalability of the filtering procedure.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix departs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) along three principal axes: objective, treatment of outliers, and technique. First, the objectives differ fundamentally. In OpenMatch Saito et al. (2021), the goal is to cope with outliers while training a standard classifier under semi-supervised learning (SSL); in our setting, by contrast, the goal is to detect OOD samples within unlabeled wild data and to train a dedicated OOD detector. Whereas SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) seek accurate classification in spite of outliers, Medix targets OOD detection itself in an open-world regime entirely devoid of labeled OOD data.

Second, the two paradigms treat outliers in diametrically opposite ways. SSL methods train a classifier robust to OOD samples by treating outliers as noise to be suppressed, thereby improving SSL performance; Medix instead explicitly detects OOD samples and actively exploits them as a training signal, constructing a more effective and reliable OOD detection system.

788 Third, the underlying techniques diverge. OpenMatch rests on consistency regularization—enforcing
789 agreement across stochastic transformations of the same input, formalized as soft constraints on the
790 classifier’s outputs that encourage smooth decision boundaries. Medix instead performs median-based
791 filtering, exploiting the statistical robustness of the median and its insensitivity to extreme values to
792 identify candidate OOD samples; its accompanying analysis derives error bounds governed by the
793 contamination effect (the OOD proportion) and the concentration effect (the sub-Gaussian behavior
794 of InD gradients), thereby guaranteeing that OOD detection error remains provably low.

795 **B Broader Impacts and Limitations**

796 As machine learning systems proliferate, robust OOD detection has become indispensable to reliable
797 model performance in real-world applications. Our median-centric framework directly advances
798 this goal, strengthening OOD handling and, with it, overall model safety. The significance of this
799 contribution extends beyond its theoretical guarantees to practical deployment across high-stakes
800 domains such as healthcare, autonomous systems, and finance: by improving OOD detection, Medix
801 addresses a central obstacle to trustworthy deployment and thereby fosters broader confidence in, and
802 adoption of, machine learning technologies. Promising avenues for future work include coupling
803 Medix with generative models for outlier synthesis and adapting it to dynamic environments in which
804 the OOD distribution drifts over time.

805 One assumption of Medix is its reliance on a mixture of unlabeled InD and OOD data that may, in
806 practice, be noisy, corrupted, or inconsistently sampled. This assumption is by no means peculiar to
807 our method; it reflects the intrinsic uncertainty of real-world deployment and constitutes a standard
808 starting point in robust learning theory. Critically, we render our assumptions explicit and verifi-
809 able—formalizing them in the main theorems and stating a precise geometric separation condition in
810 Remark 4.3—and we relax them substantially in Theorem C.3, which dispenses with sub-Gaussian
811 tails and establishes the resilience of our framework under considerably weaker distributional as-
812 sumptions. Taken together, these results demonstrate that the simplifying assumptions of Medix are
813 at once interpretable and relaxable, constituting a principled foundation for future advances in OOD
814 detection under realistic noise.

815 C Proof of Theorem

816 For clarity and consistency across the theoretical and empirical sections, Table 7 consolidates the key
 817 notation employed throughout the paper, covering gradient-related quantities, sample sizes, filtering
 818 terms, and the parameters that enter our concentration bounds. Unless stated otherwise, we denote
 819 vectors in boldface, write $\mathcal{X}$ for the input space, and compute all gradients with respect to the fixed
 820 pretrained model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

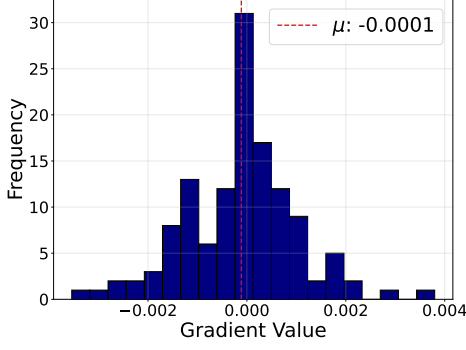
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

821 C.1 Bound on the Inlier Misclassification Rate

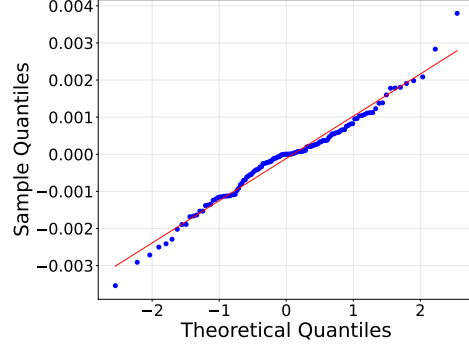
822 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 823 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 824 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 825 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

826 *Proof.* The argument proceeds in two steps. In **Step 1** we establish a high-probability concentration
 827 bound for the gradients of InD samples about their mean $\bar{\mathbf{v}}_{\text{in}}$, designating as a *good inlier* any InD
 828 sample whose gradient falls within this high-probability region. In **Step 2** we invoke a “swapping”
 829 argument on the EWM objective to show that excluding more than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of
 830 good inliers—where η denotes the fraction of InD samples with gradients outside the high-probability
 831 region—would contradict the optimality of the minimizer. Combining the two steps yields the stated
 832 upper bound on the inlier misclassification rate ERR_{in} .



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Empirical evidence that InD sample gradients are coordinate-wise sub-Gaussian. (Left) The histogram of gradient values (CIFAR-100 InD data) concentrates tightly about the mean with light tails, exactly as expected under sub-Gaussianity. (Right) The Q-Q plot of empirical InD gradient quantiles against theoretical Gaussian quantiles aligns closely with the reference line, confirming sub-Gaussian behavior. Jointly, these two panels directly justify the sub-Gaussian assumption underlying Theorems 4.1 and 4.2.

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ) of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

848 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

849 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

850 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
851 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

852 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

853 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

854 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

855 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

856 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
857 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
858 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
859 called *good inliers*.

860 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 861 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

862 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 863 thereby contradicting the optimality of $\mathcal{S}^*$.

864 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 865 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

866 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

867 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

868 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

869 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 870 available for swapping in.

871 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 872 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

873 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

874 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

875 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

876 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

877 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 878 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

879 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

880 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

881 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

882 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

883 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

884 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
885 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

886 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
887 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
888 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

889 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

890 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
891 increase it, so the median distance from μ_j does not increase and possibly decreases.

892 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

893 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

894 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
895 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

896 □

897 C.2 Bound on the Outlier Retention Rate

898 We now establish a non-asymptotic upper bound on the fraction of OOD points retained in the subset
899 returned by the MEDIX filter. The result parallels the inlier misclassification bound (Appendix C.1)
900 but, rather than relying on inlier concentration, exploits a geometric separation assumption between
901 the OOD and InD gradient means—the natural complement that closes the two-sided guarantee.

902 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 903 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

904 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

905 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

906 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

907 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

908 *Proof. Step 1: Concentration of OOD Gradients.*

909 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 910 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

911 has probability at most $p_{\text{out}}(\varepsilon)$.

912 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

913 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

914 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

915 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

916 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

917 which holds with probability at least $1 - \delta$.

918 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

919 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

940 **Differences with the original bound.** The original, sub-Gaussian bound carries a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

941 that vanishes exponentially in the dimension d , rendering the bound exceptionally tight for large d .
 942 The present bound, by contrast, invokes only finite fourth moments and therefore exhibits the milder
 943 decay term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2},$$

944 which decays as $O(1/d)$ and is correspondingly looser at large d . The trade-off is thus fully explicit:
 945 the original bound requires sub-Gaussian tails, whereas the present one tolerates substantially heavier
 946 tails under nothing more than a finite-fourth-moment condition. Crucially, both bounds retain the
 947 identical finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

948 which shrinks as m_{in} grows irrespective of the tail assumption—so the core robustness guarantee
 949 persists intact across both regimes.

950 *Proof.* The proof mirrors the structure of the original Theorem 4.1 and again comprises two steps:
 951 (1) bounding the fraction of “bad” InD samples (those far from ∇_{in}) via concentration, now without
 952 invoking sub-Gaussianity, and (2) bounding the inlier misclassification rate through a swapping
 953 argument. The sole substantive modification is that the sub-Gaussian tail bound is replaced by a
 954 moment-based bound derived from the finite-fourth-moment assumption—precisely the substitution
 955 that enables the generalization to heavier-tailed gradient distributions.

956 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

957 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 958 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

959 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 960 S_{wild} .

961 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 962 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

963 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

964 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

965 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

966 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

967 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

968 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

969 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

970 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
971 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

972 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
973 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

974 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

975 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

976 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

977 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 978 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

979 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 980 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

981 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 982 thereby contradicting the optimality of $\mathcal{S}^*$.

983 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 984 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

985 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

986 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

987 Define the swap size as

$$m'_\epsilon := \min \{m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

988 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 989 available for swapping in.

990 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 991 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

992 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

993 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

994 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

995 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

996 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 997 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

998 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

999 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

1000 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

1001 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

1002 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

1003 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 1004 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

1005 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 1006 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 1007 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

1008 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

1009 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 1010 increase it, so the median distance from μ_j does not increase and possibly decreases.

1011 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

1012 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

1013 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 1014 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

1015 □